Let us consider the formation of a metal complex $ML_6$ in aqueous medium.(Charge on the metal ion is ignored) complex formation may occur in single step or step by step.

If ligands added to the metal ion in single step, then

$$[MS_6] + 6\,L \rightleftharpoons [ML_6] + 6\,S$$

$$\beta_{overall} = \frac{[ML_6]\,[S]^6}{[MS_6]\,[L]^6}$$

$\beta_{overall}$ is called as overall stability constant. As solvent is present in large excess, its concentration in the above equation can be ignored.

$$\therefore\ \beta_{overall} = \frac{[ML_6]}{[MS_6]\,[L]^6}$$

If these six ligands are added to the metal ion one by one, then the formation of complex $[ML_6]$ can be supposed to take place through six different steps as shown below. Generally step wise stability constants are represented by the symbol k.

$$[MS_6] + L \rightleftharpoons [MS_5L] + S \qquad k_1 = \frac{[MS_5L]}{[MS_6][L]}$$

$$[MS_5L] + L \rightleftharpoons [MS_4L_2] + S \qquad k_2 = \frac{[MS_4L_2]}{[MS_5L][L]}$$

$$[MS_4L_2] + L \rightleftharpoons [MS_3L_3] + S \qquad k_3 = \frac{[MS_3L_3]}{[MS_4L_2][L]}$$

$$[MS_3L_3] + L \rightleftharpoons [MS_2L_4] + S \qquad k_4 = \frac{[MS_2L_4]}{[MS_3L_3][L]}$$

$$[MS_2L_4] + L \rightleftharpoons [MSL_5] + S \qquad k_5 = \frac{[MSL_5]}{[MS_2L_4][L]}$$

$$[MSL_5] + L \rightleftharpoons [ML_6] + S \qquad k_6 = \frac{[ML_6]}{[MSL_5][L]}$$

In the above equilibrium, the values $k_1, k_2, k_3, k_4, k_5$ and $k_6$ are called step wise stability constants. By carrying out small a mathematical manipulation, we can show that overall stability constant $\beta$ is the product of all step wise stability constants $k_1, k_2, k_3, k_4, k_5$ and $k_6$.

$$\beta = k_1 \times k_2 \times k_3 \times k_4 \times k_5 \times k_6$$

On taking logarithm both sides

$$\log(\beta) = \log(k_1) + \log(k_2) + \log(k_3) + \log(k_4) + \log(k_5) + \log(k_6)$$



XII U5 Coordination jagan.indd 165 04-01-2021 13:24:25